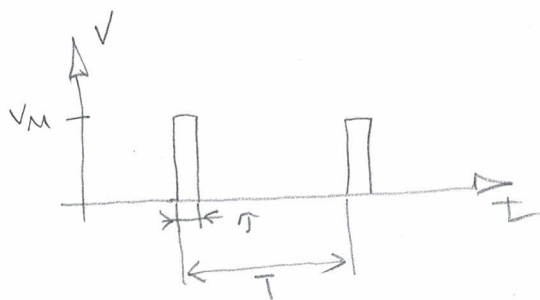


4.32)



BERÄKNA $V_{\text{medel}} / V_{\text{rms}}$ för vågformen ovan.

Lösning:

$$\begin{cases} V_{\text{medel}} = \frac{1}{T} \int_0^T V(t) dt = \frac{1}{T} T V_m \\ V_{\text{rms}}^2 = \frac{1}{T} \int_0^T V^2(t) dt = \frac{1}{T} \int_0^T V_m^2 dt = \frac{1}{T} V_m^2 T \end{cases}$$

KVOTEN BLIR

$$\frac{V_{\text{medel}}}{V_{\text{rms}}} = \frac{\frac{1}{T} T V_m}{\sqrt{\frac{1}{T} T} V_m} = \sqrt{\frac{T}{T}}$$

SVAR :

$$\frac{V_{\text{medel}}}{V_{\text{rms}}} = \sqrt{\frac{T}{T}}$$